SIMATS ENGINEERING

ASSESSMENT TOOL 1 (High)- Analytical Problem Solving

COURSE NAME: Data Structures **COURSE CODE:CSA0304**

COURSE OUTCOME COVERED

CO2 : Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques. (BTL3, BTL4)

Assessment Tool Weightage: **30%**

QUESTIONS:

1. Perform the following operations in LIFO data structure having an array size of 5. Push(45), Push(22), Push(11), Push(77), Pop(), Push(44), Push(33), Pop(), Pop(),Pop(). Print the peak element of the data structure and its index position. With a neat sketch explained in detail.

2. How do you implement Queue data structure using Single Linked List?. With a neat diagram discussing Enqueue and Dequeue operations using SLL.

3. Perform the following operations in Circular Queue having a size of 5. Enqueue(20), Enqueue(10), Enqueue(30), Dequeue(), Enqueue(80), Dequeue(), Enqueue(90), Enqueue(100), Dequeue(), Dequeue(), Enqueue(60), Enqueue(90). Finally display the front and rear elements with its positions. Discuss in detail.

4. Implement Binary searching Algorithm for searching an element and its position having the following 10 elements. Elements are 66, 111, 212, 323, 432, 543, 643, 649, 777, 888. Case 1: Key element = 111, Case 2: Key element=888, Case 3: Key element = 999. Analyse the concept in detail.

5. Pictorially insert an element at 3^{rd} position in a doubly linked list, having 6 elements in the list and delete the element at 2^{nd} position from the list. Implement and analyse it.

6. Perform the following operations using stack. Assume the size of the stack is 5 and have a value of 22, 55, 33, 66, 88 in the stack from 0 position to size-1.

Now perform the following operations: 1) Invert the elements in the stack, 2) Pop(), 3) Pop(), 3) Pop(), 4) Push(90), 5) Push(36), 6) Push(11), 7) Push(88), 8) Pop(), 9) Pop(). Draw the diagram of the stack and illustrate the above operations and identify where the top is?

7. Convert the following infix expression into a post fix expression. B - (C/D + (E % F* G) / H) * J. Also, evaluate the given postfix expressions. **a)** 9 3 8 4 /*+ b) 6 2 + 6 3 /*

8. Illustrate the queue operation using following function calls of size = 5. Enqueue(21), Enqueue(67), Enqueue(89), Dequeue(), Enqueue(15), Enqueue(30), Enqueue(52), Dequeue(), Dequeue(), Dequeue(),Dequeue().

9. Identify if the given expressions are balanced or not. (i) [9 *(k+8 -{ 6 +e} + 77) / 100, (ii) (((a*b) +[u/2] –d).

10. Given a stack with repeated elements, design an approach to compress it (value & frequency) while maintaining stack operations.
    - Apply stack ADT modifications.
    - Analyze trade-off between time and space

11. Insert the following elements 100, 200, 300, 400, 500 into the linked stack and pop the elements 100 and 200 from the linked stack. Show the linked structure before and after linked stack operations.

12. Design a system that dynamically chooses between linear search and binary search based on:
    - Data size
    - Sortedness
    - Frequency of queries
    - Analyze decision criteria mathematically

13. Construct an efficient system using two stacks within a single array structure to maximize memory utilization. Compare fixed partitioning and dynamic allocation strategies, and analyze how each approach affects overflow conditions, space efficiency, and operational complexity under varying workloads.

14. A queue-based ticketing system serves customers in the order of arrival, but VIP customers must be prioritized occasionally. Analyze how modifying a simple queue into a priority-based queue affects fairness, waiting time, and

computational complexity. Compare both systems using analytical reasoning.

15. Customers are standing in a line before a Banker Casher Count to perform bank transactions. What data structure will be suitable to perform customer transactions. Give a suitable data structure diagram to serve the customer without bias.

16. Insert the following elements 150, 250, 350, 450, 550 into the double linked list and delete the elements 150 and 250 from the double linked list. Show the linked structure using a node diagram before and after delete operations and explain the processing steps with a suitable procedure.

17. Ten documents are requested for printing. Illustrate the mechanism of printer operations. Consider the document number as 1003, 1008, 1010, 1005, 1009, 1004 . Identify the suitable data structure and draw the diagram to print the documents.

18. The university announced the selected candidates register number for placement training. The student XXX, Register Number 20142010 wishes to check whether his name is listed or not. The list is not sorted in any order. Identify the searching technique that can be applied and explain the searching steps with suitable procedure. List includes 20142015, 20142033, 20142011, 20142017, 20142010, 20142056, 20142003

19. Insert the following elements 150, 250, 350, 450, 550 into the double linked list and delete the elements 150 and 250 from the double linked list. Show the linked structure using a node diagram before and after delete operations.

20. Mr. XYZ booked a ticket on Pandian Express. He reached the station and was looking for his PNR number in the chart. Call out binary search procedure to search the PNR number 4056786. The ticket numbers are 4056779, 4056780, 4056781, 4056782, 4056783, 4056784, 4056785, 4056786, 4056787, 4056788, 4056789.

21. Convert the expression ((a + b) + c * (d + e) + f) * (g + h) to 1) Prefix expression , 2) Postfix expression.

22. Perform the binary search in an array of n elements using a recursion function, in which a function has to call by itself. Use these elements given in sorted order and assume 88 as the key element. [23, 33, 43, 48, 53, 58, 61, 64, 68, 73, 88]

23. How a queue may be maintained by a circular array with MAXSIZE = 6. Observe that queue always occupies consecutive locations except when it occupies locations at the beginning and at the end of the array. If the queue is viewed as a circular array, this means that it still occupies consecutive locations. The queue is empty when count =0; Perform the following Enqueue and Dequeue operations. a) Initially QUEUE is empty, b) A,B,C are Enqueued, c) A is Dequeued, d) D,E,F are Enqueued, e) B & C are Dequeued, f) G is Enqueued , g) D & E are Dequeued, h) H & I are Enqueued, i) F is dequeued, j) G & H are dequeued k) I is Dequeued. Demonstrate with a neat sketch of Front, Rear and count in a queue and circular queue.

24. Show the linked list configuration resulting from each series of list operations using the List ADT. Assume that lists L1 and L2 are empty at the beginning of each series. Show where the current position is in the list. (a) L1.append(10); L1.append(20); L1.append(15); (b) L2.append(10); L2.append(20); L2.append(15); L2.moveToStart(); L2.insert(39); L2.next(); L2.insert(12);

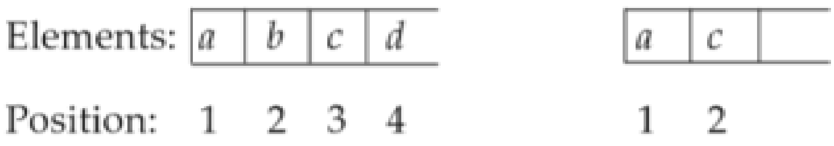
25. Design a program to check the validity of a given expression containing parentheses, square brackets, and curly braces. Implement a stack data structure to validate if the brackets in the expression are balanced and properly nested. Provide a function that returns 'true' if the expression is valid, and 'false' otherwise

26. A palindrome is a string that reads the same forwards as backwards. Using only a fixed number of stacks and queues, the stack and queue ADT functions, and a fixed number of int and char variables, write an algorithm to determine if a string is a palindrome. Assume that the string is read from standard input one character at a time. The algorithm should output true or false as appropriate.

27. Given a singly linked list L: L1-> L2->L3...-> Ln−1->Ln, reorder and print it to: L1-> Ln ->L2 -> Ln−1.

28. Insert the following elements 150, 250, 350, 450, 550 into the double linked list and delete the elements 150 and 250 from the double linked list. Show the linked structure using a node diagram before and after delete operations.

29. Implement a stack S of n elements using arrays. Write functions to perform PUSH and POP operations. Implement queries using the push and pop functions to Retrieve the mth element of the stack S from the top (m<n), leaving the stack without its top m - 1 elements. Retain only the elements in the odd position of the stack and pop out all even positioned elements.



30. Consider a linked list representing the sales of different products in a store for a week. Each node of the linked list contains the product ID and the number of units sold on a specific day. Implement a linked list data structure and provide functions to: Insert new sales data for a given product on a specific day. Calculate the total sales for each product at the end of the week. Find the product with the highest sales and its corresponding day. Display the sales data for a specific product for the entire week. Use your own data for product.

31. Design a calculator program that evaluates postfix expressions using a stack data structure. The calculator should take a postfix expression as input and return the result of the expression. Given the postfix expression: 5 3 8 * + 4 - Implement the calculator program using a stack data structure to handle the postfix evaluation efficiently."

32. Write a program to find the sum of all the elements in a single linked list having the elements of 34->77->22->55->66->11-> in the list.

33. Find the difference between the minimum and maximum values in a single linked list having 34->77->22->55->66->11-> in the list.

34. The double linked list has two links in each node as “prev”, “data” and “next”. Display the elements from DLL in forward and reverse order and find the average of those ‘n’ elements.

35. Print all the odd and even counts present in the single linked list having 10 elements in the list 77->88->22->11->66->27->55->65->18->10->

36. Search linearly a key elements K=20 present in the list of single lined list. If so print “YES” else print “NO”.

37. Assume, a single character is present in the data field of DLL with 6 nodes as S<=>I<=>M<=>A<=>T<=>S. Write a function to print the same in reverse order.

38. Assume a single character is present in the data field of a Single Linked List with 6 nodes as S- > I -> M -> A -> T -> S. Write a function to print the same in reverse order using stack data structure.

39. Write an algorithm called Find-Largest that finds the largest number in an array using a divide-and-conquer strategy.

40. Print all the prime integers present in the Single linked list having a list of 24->21->57->31->12->47->7->38->.

Code of Conduct:

*I **Kaviya.G(192524281)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

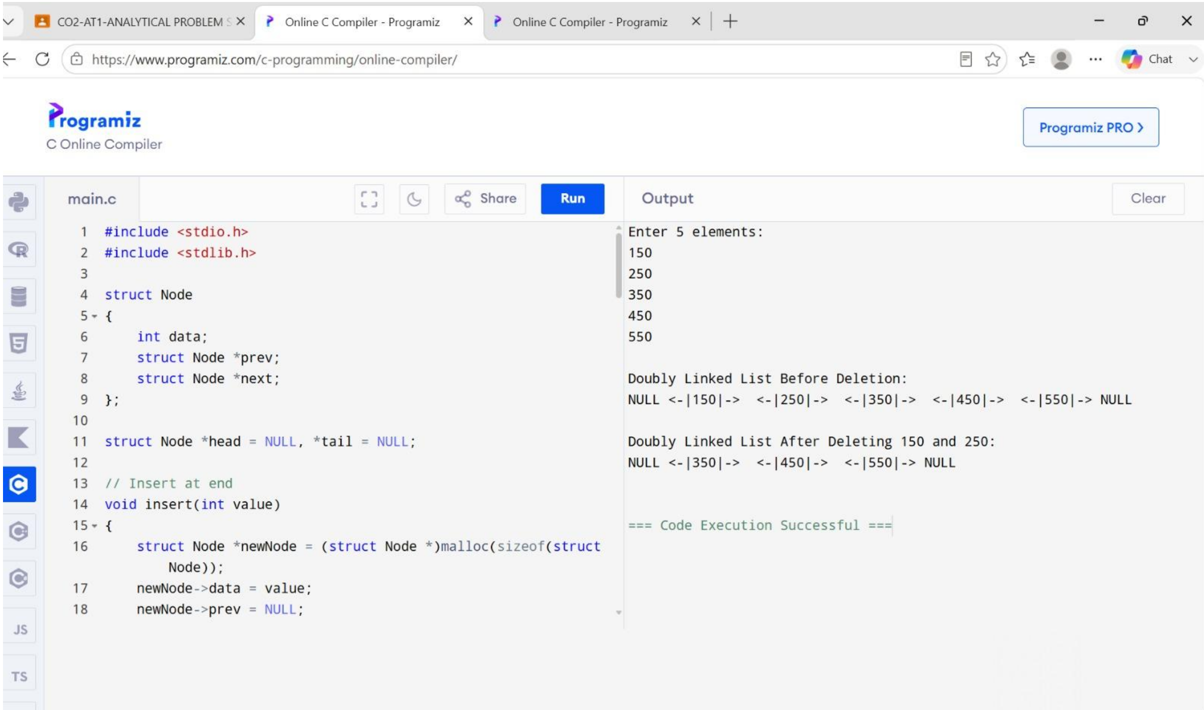
Signature of the student: **Kaviya.G**

RUBRICS FOR CALCULATION:

Numerical Problems

Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

19.Insert the following elements 150, 250, 350, 450, 550 into the double linked list and delete the elements 150 and 250 from the double linked list. Show the linked structure using a node diagram before and after delete operations.



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